

Reward allocation in Energy Communities by size, composition and prosumers penetration

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Abstract—The energy transition calls for citizen-driven investments in distributed renewables, supported by policies that enable Energy Communities. However, the viability of such initiatives hinges on the fair and transparent allocation of collective revenues in agreement to the community size and composition. Classical game-theoretic allocations, while offering fairness and coalition stability, are often too complex and opaque for real-world implementation. This paper investigates the fair allocation of rewards in Energy Communities depending on community size, composition and prosumers penetration, to infer policy and technical insights. We adopt the game-theoretic reward allocation Variance Least Core that is unique and stable. To break down computational requirements, we adopt a row-generation technique in a Mixed-Integer Linear programming framework that breaks down the typical computational requirements of game-theoretic reward allocation. The methodology is tested across 64 case studies varying in size, prosumer share, and user diversity. The results confirm that the proposed algorithm delivers fair, reproducible, and regulator-ready reward allocations for large-scale energy communities—thus making advanced game-theoretic principles practically usable within open-source tools.

Index Terms—Renewable energy; Energy sharing; Renewable energy community; Game theory; Reward

I. INTRODUCTION

A. Motivation

The energy transition requires a wholesale rethink of how we plan and operate power systems. Because renewables are intermittent, balancing the grid now depends on decentralised generation, distributed storage, and active prosumer participation. Unlocking those investments demands clear policy and market rules that reward citizens for supporting the network [1], [2]. European Union and national policies support Energy Communities (ECs) through shared energy incentives, but leave internal reward allocation to members, raising fairness concerns [3]. For these reasons, we deemed useful investigating the role of optimal reward allocation for ECs.

B. Literature analysis

In ECs, users of the power grid becomes members and are allowed to do cojoint investment and operation of energy assets. Each member pays its own bill, but a public authority awards the community with a premium quantity for every kilowatt-hour being produced by a member and consumed by

another. The total premium is then used by the aggregate as it best see fit in agreement to the membership agreement. In ECs, aggregators that coordinate the overall operation of the system are expected [4], which facilitates achieving the maximum profitability of the energy investment, usually measured with net-present value [5]. Besides accelerating decarbonisation, proven system-level benefits include extra reserves [6], de-congestion [7], higher renewable penetration [8] and net social welfare gains [9]. As member of ECs rarely have appropriate technical expertise, ECs often delegate the planning and operation of the system to an aggregator that has appropriate technical background [10]. The planning usually involves optimization techniques, usually based on Mixed-Integer Linear Programming [11], [12]. Given the inclusive nature of ECs [3], approaches typically focus on maximising community benefit [13] or welfare [14], [15] but sidestep fair profit sharing; the aggregator's own stable remuneration is seldom tackled [4].

To ensure the optimal planning is actually achieved in real-time operation [11], appropriate reward sharing techniques that ensure meeting the EC goal needs to be designed. Indeed, fair allocation preserves the attractiveness of the EC in acquiring and retaining members, which means that members proactively act towards the common goal. On this regard, cooperative game theory offers a number of well established techniques to measure fairness. Game-theoretic schemes like Shapley Value, Least Core, Core, and Nucleolus offer fairness and stability guarantees, but their complexity hinders practical adoption in large ECs [4], [8]. They typically involve a combinatorial number of iterations that becomes intractable beyond 10-20 users as proven in [16] where 2 months of computations were needed to compute Shapley Value for 20 users. However, the recent VarianceLeastCore technique—being a subset of ConvexLeastCore—has demonstrated its applicability for large communities thanks to mathematical decomposition [16]. Nevertheless, the methodology has never been used to investigate the optimal reward allocation, considering their market power in communities of increasing number of members, different penetration of prosumers and configuration types.

C. Contributions and organization of the paper

In this paper we investigate the role of size of the community, prosumers penetration and technical planning into the fair reward allocation of ECs by adopting game theory and mathematical decomposition in the EC optimal planning. Based on the procedure in [16], we developed the procedure and applied it to a large number of realistic case studies of communities between 10 and 50 members. Each community may have a number of prosumers between 0 and 100% and we considered two types of energy assets configurations to comprehensively investigate on the optimal fair reward allocation of members to infer policy and technical decision making.

The remainder of the paper is organized as follows. Section II describes the EC and its mathematical optimization problem. Section III and Section IV denotes the game-theoretic function and methodology adopted in this study. The case study and results are reported in Section V and VI, respectively. Finally, conclusions are drawn.

II. OPTIMAL PLANNING BY ENERGYCOMMUNITY.JL

To evaluate the optimal planning of an EC we developed our methodology leveraging on the Julia package *EnergyCommunity.jl*. The tool allows the techno-economic planning of an EC under multiple configurations: the Non-Cooperative (NC) and Cooperative (CO). In the CO configuration, by means of an aggregator, the users cooperate to achieve the maximum Social Welfare, or total Net Present Value, for the aggregate accounting for the reward proportional to the energy shared among members. In the NC configuration, which represents the base case, no EC is in place and each user maximizes its own Net Present Value with no additional reward for energy shared among members. Each member may install energy assets behind its Point-of-Delivery.

A. Objective and shared energy

In a CO EC, the overall goal is to maximize the so-called social welfare $SW^{CO}(K)$ of the community K , which includes the Net Present Value (NPV) of each member j and the total reward R_y^{SH} allocated to the community, as detailed in (1), where r is the annual discount rate applied to future revenues, and y represents the year in the investment horizon Y . The total reward annually awarded to an EC is formulated in (2), where π_t^{SH} is the regulated unitary reward and P_t^{SH} is the shared energy virtually net-metered. P_t^{SH} is defined as the minimum between the hourly overall production U^+ and consumption U^- , as modelled in (3).

$$SW^{CO}(K) = \sum_{j \in K} NPV_j + \sum_{y \in Y} \frac{R_y^{SH}}{(1+r)^y} \quad (1)$$

$$R_y^{SH} = \sum_t \pi_t^{SH} P_t^{SH} \quad (2)$$

$$P_t^{SH} = \min \left\{ \sum_{j \in K} P_{j,t}^{U+}, \sum_{j \in K} P_{j,t}^{U-} \right\} \quad \forall t \quad (3)$$

B. Coordinated EC problem (CO)

The mathematical problem for the coordinated EC is shown in (4), while constants c_j and l_j represent the cost coefficients discussed in [16]. The shared power s is constrained to be lower than or equal to the total energy production and consumption, by using matrices $D^\pm$, through the identity $P_j^{U\pm} = D^\pm u_j$; $\delta > 0$ represents the weighted reward for every unit of shared power.

$$SW^{CO}(K) = \max_{\substack{s \leq \sum_{j \in K} D^+ u_j \\ s \leq \sum_{j \in K} D^- u_j}} \sum_{j \in K} (c_j^T u_j + l_j) + \delta^T s \quad (4)$$

The shared energy is computed as the minimum between the energy fed and consumed conjointly within the EC as indicated in (5) and (6).

$$E_t^{SH} \leq P_{j,t}^{M+} \Delta \quad (5)$$

$$E_t^{SH} \leq P_{j,t}^{M-} \Delta \quad (6)$$

Given the modeling mentioned above, the function $v(J)$ is calculated in (7) as the difference between the CO and NC configurations for a given set J of members of the community.

$$v(J) = NPV^{CO}(J) - NPV^{NC}(J) \quad (7)$$

III. VARIANCE LEAST CORE REWARD ALLOCATION

Reward allocation in cooperative game theory refers to the field of study aiming to fairly distribute a total utility $v(\cdot)$ among a set of members (or players) I . The function $v(K)$ denotes the total utility generated by an arbitrary coalition $K \subseteq I$ of the players I , that shall be distributed. In the case of ECs and for the purpose of this paper, the utility is the difference between the objective value obtained in the CO EC versus the NC configuration, which is the base case. In mathematical terms, eq. (8) describes the utility function $v(K)$ for the EC problem described in Section II for any coalition $K \subseteq I$, where I denotes all members of the community including the Aggregator (A) and $K_A = K \setminus \{A\}$ denotes the set K without the aggregator.

$$v(K) = \begin{cases} SW^{CO}(K_A) - SW^{NC}(K_A) & \text{if } A \in K, \\ SW^{ANC}(K) - SW^{NC}(K) & \text{if } A \notin K, \end{cases} \quad (8)$$

Each user i perceives an allocation Δ_i and the total shall match the available reward such that $\sum_{i \in I} \Delta_i = v(I)$. Therefore, the set of possible allocations Δ can be denoted by the set $\mathcal{B}$, defined as in (9). This definition allows to ease the mathematical reformulation in the following.

$$\mathcal{B} = \{ \Delta \in \mathbb{R}_+^{|I|} : \sum_{i \in I} \Delta_i = v(I) \} \quad (9)$$

Once the allocation vector Δ is defined, then the improved objective function NPV_i^F of each user is calculated as in (10), where NPV_i^{NC} denotes the objective function of the base case NC case described in the previous section.

$$NPV_i^F = NPV_i^{NC} + \Delta_i. \quad (10)$$

To ease the definition of the game-theoretic function, it is useful to define the concept of surplus $\sigma(K, \Delta)$ in (11), which when positive denotes that users $i \in K$ are better off in the community.

$$\sigma(K, \Delta) = \sum_{i \in K} \Delta_i - v(K) \quad (11)$$

A. Least Core

To best clarify the Variance Least Core (VLC), we introduce the set Least Core ($\mathcal{LC}$) that is the set of allocations defined in (12) that guarantees that no coalition K is worse off within the coalition by the quantity θ^{LC} , calculated in (13). When $\theta^{LC} > 0$, each user has a strictly positive benefit in participating and remaining in the community. This means that no user has the incentive from leaving the community, hence strengthening coalition stability. When $\theta^{LC} = 0$, there may be some users that are better off in the community than outside, the set is also known as *Core*.

$$\mathcal{LC}(I, v) = \{\Delta \in \mathcal{B} : \sigma(K, \Delta) \geq \theta^{LC} \quad \forall K \subset I\} \quad (12)$$

$$\begin{aligned} \theta^{LC} = \max \quad & \theta \\ \text{s.t.} \quad & \sigma(K, \Delta) \geq \theta \quad \forall K \subset I \\ & \Delta \in \mathcal{B} \end{aligned} \quad (13)$$

It is worth noting $\mathcal{LC}$ is a set and in general multiple allocations exist. Moreover, it is worth noting that the number of constraints in (12) is combinatorially increasing with respect to the size of the community $|I|$, which can lead to computational challenges.

B. Variance Least Core

To ensure uniqueness of the solution, the authors in [4], [16] proposed to identify in the Least Core (or Core) a specific allocation that minimizes the squared distance from the uniform allocation, as defined in (14).

$$\Delta^{VLC} = \arg \min \left\{ \sum_{i \in I} \left[\Delta_i - \frac{v(I)}{|I|} \right]^2 : \Delta \in \mathcal{LC}(I, v) \right\} \quad (14)$$

This formulation ensures uniqueness, yet it is still a combinatorial problem that can lead to computational problems.

C. Mathematical Decomposition

1) *Description*: To break down the computational requirements and adopt VarianceLeastCore, we used the row-generation technique denoted in [16] that adopts an iterative approach between a Master problem and a Separation problem to both identify Least Core and VLC. The procedure for identifying the VLC is as follows:

- Optimize Master Problem (15) to identify a tentative allocation Δ^M that meets the least core.
- Optimize Separation Problem (17) to identify whether the allocation Δ^M belongs to the Least Core.
- If the surplus matches least core, the procedure converges; otherwise the coalition that is worse off is added to the master problem and the procedure repeats

A similar procedure is adopted to identify the surplus θ^{LC} , which adopts a slightly different Master Problem in (16).

2) *Master Problem*: The Master Problem in (15), where f denotes the VLC, is equivalent to (12) but insisting on the subset Γ that starts with few coalitions and grows in each iteration.

$$\max \{ f(\Delta) : \Delta \in \mathcal{B}, \sigma(K, \Delta) \geq \theta^{LC} \quad \forall K \in \Gamma \}, \quad (15)$$

To calculate θ^{LC} , the master problem in (12) is instead adopted.

$$\max \{ \theta : \Delta \in \mathcal{B}, \sigma(K, \Delta) \geq \theta \quad \forall K \in \Gamma \}, \quad (16)$$

3) *Separation Problem*: The goal of the Separation Problem in (17) is to identify if the allocation Δ^M generated by the Master Problem belongs to the Least Core. This is achieved by creating an optimization problem that identifies the coalition that has the lowest surplus given the allocation Δ^M generated by the Master Problem. By using the binary variables z to map a generic coalition K , such that if $z_i = 1$, user i belongs to K , it is possible to represent the problem using (17).

$$\begin{aligned} \min_z \quad & \sum_{i \in I} \Delta_i^M z_i - v(z) \\ \text{s.t.} \quad & 1 \leq \sum_{i \in I} z_i \leq |I| - 1 \\ & z \in \{0, 1\}^{|I|} \end{aligned} \quad (17)$$

IV. METHODOLOGY

To investigate the fair reward allocation by community size, compositions and technical part, in this study we implemented the following methodology:

- 1) Identification of representative ECs: this step aims to identify key characteristics of ECs that may influence the fair reward allocation
- 2) Preparation of the representative communities and the set of parameters to investigate
- 3) Optimal planning and reward allocation adopting the methodology in sections Section II and Section III
- 4) Post-processing and results analysis

Based on the literature [4], [5], [8], [16], it emerges that critical characteristics affecting reward allocation is (a) community size, (b) members composition, (c) the technical assets and location among the members and (d) the number of prosumers as a share of the total. These considerations guided the definition of the representative communities to be investigated in the case study.

V. CASE STUDY

We considered a representative case study for Italy, using the reference case provided by [17] for comparability with the literature. As discussed previously, and based on the procedure defined in Section IV we developed a number of configurations exploring on community size, prosumer penetration, and technology penetration. We considered community sizes from 10 to 50 members, which are representative of Italian case studies; the communities larger than 10 members are generated by replicating the original 10 members of [17]. Moreover, to stress the role of prosumers, we considered a penetration spanning from 10 to 100%. Finally, we considered different availability of technical components: the *mixed* configuration

TABLE I
SUMMARY OF CASE STUDY SCENARIOS

Users #	Prosumers %	Prosumer Types	Noise %	Simulations #
10	20, 50, 70, 100	Mixed	0	8
30	20, 50, 70, 100	Mixed	0	8
50	20, 50, 70, 100	Mixed	0	8
10	20, 50, 70, 100	Identical	0	8
30	20, 50, 70, 100	Identical	0	8
10	20, 70	Identical	10, 20	8
30	20, 70	Identical	10, 20	8
10	10	Identical-inf	10, 20	4
30	10	Identical-inf	10, 20	4

allows for heterogeneous availability of wind, pv and battery technologies across members, contrary to the *identical* configuration. This results in 64 main simulations, 32 NC and 32 CO allowing a systematic assessment of cooperation in EC under varying conditions. *Identical-inf* and *Noise* leave no limits on the installed capacity and an increase of 10 and 20% of users load.

VI. RESULTS

The results of the test procedure are shown in Figures from 1 to 2, and are discussed in the following.

A. Objective function comparison

CO optimisation boosts NPV relative to NC, with the largest gains in 50-member groups at 70–100 % prosumer penetration. For each community size, the best-performing scenarios are all CO, while the NC configurations consistently yield the lowest outcomes.

The smaller scenarios show more similar NPVs, since less local energy is generated and the dependency on shared optimization is lower. These results emphasize the need to account for forecast uncertainty when designing ECs, especially in bigger communities, where individual behavior has a larger impact on collective performance.

The simplest Mixed and high prosumer shares achieve the best performances. This implies that to have a fair redistribution according to the theory, users must not automatically receive a reward, but the incentive mechanism shall be based on measurable quantities that relate to the shared energy.

B. Minimum Surplus

The analysis of the minimum surplus in Figure 1 highlights the robustness of the VLC allocation in maintaining coalition stability. Most communities exhibit strictly positive minimum surplus values, confirming that no subset of users is incentivized to leave the coalition. Interestingly, scenarios with 10 users and mixed configurations show higher surplus, indicating more uniform benefit distribution under simpler technical conditions. Conversely, larger communities (30 and 50 users) present a wider changes of minimum surplus values.

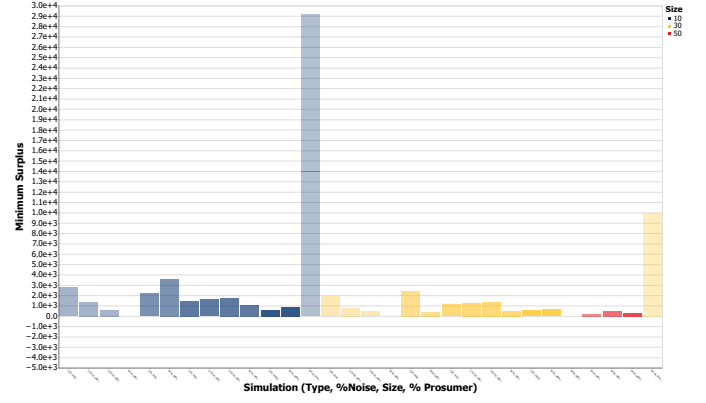


Fig. 1. Minimum surplus for all scenario

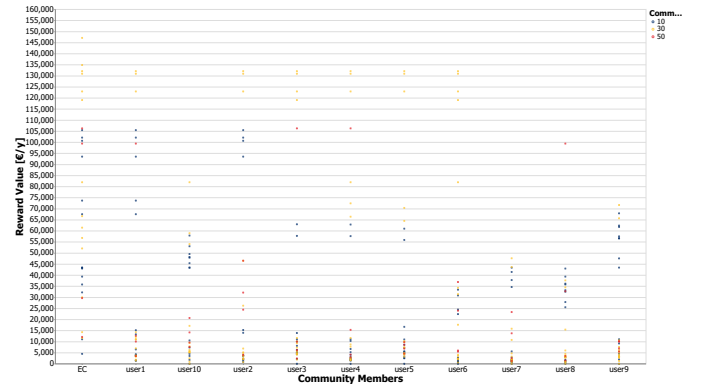


Fig. 2. Reward Distribution for EC and First 10 Users Across All Scenarios

C. Reward distribution

Reward distribution results shown in Figure 2 reveal clear patterns tied to community size and internal heterogeneity. The reward allocation is visibly more concentrated in scenarios with smaller groups (10 users), where individual contributions are more distinct and impactful. As community size increases, the distribution appears flatter, with diminished individual variability. This is coherent with the assumption of replicated user types in larger communities, as error bars are narrow or negligible. Notably, EC-level rewards tend to dominate over those allocated to individual users, reaffirming the centrality of the aggregator and collective benefit in large-scale configurations. The color-coded bars by size also show that the largest rewards per user are typically achieved in 10-user ECs, aligning with prior observations that smaller, high-penetration communities offer clearer marginal benefits under cooperative schemes.

D. Convergence characteristics

These results shown in Figure 3 confirm the potential ability of the proposed algorithm to scale in size, making game-theoretical allocation approaches feasible for large ECs. This is helped by the fact that large ECs are often composed by a small share of prosumers, which significantly reduces the design decisions and therefore the expected computational ef-

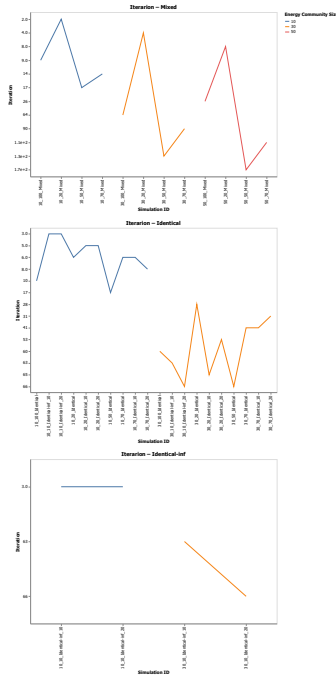


Fig. 3. Iterations required for all simulation

forts, and that performances of MILP solvers are continuously improving.

The marginal reduction decreases the larger the community. As the LC changes, the users' relative reward allocation changes, to reflect the different market power within the community. For simplicity, as for 20-member EC or larger the users are identically replicated, error bars depict the maximum and minimum benefit allocation between the same member types.

VII. CONCLUSIONS

This paper proposes a novel, stable, and reproducible reward allocation methodology for ECs, leveraging the VLC within a game-theoretic framework. The approach ensures fairness and coalition stability by preventing any member from being worse off within the community than outside. A key innovation is the use of a row-generation algorithm, which significantly reduces the computational burden of classical cooperative allocation methods. Results across 64 simulations show that the method is scalable to large ECs, while delivering allocations that are fair, unique, and regulator-ready. Tested on a realistic Italian case study, the methodology proves adaptable to different sizes, prosumer shares, and technical configurations. It provides a robust foundation for implementing measurable, energy-based reward schemes and highlights the limitations of uniform or demand-only distributions in ensuring fairness and stability.

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